

The influence of Analyst Sentiment on the VIX Index

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Abstract

We find that the Chicago Board Options Exchange Volatility Index (VIX) can be predicted by using textual information retrieved from institutional research analysts' US macro reports. Often described as the “fear” index, the VIX reflects investors' expectation of market volatility. Our analysis reveals a strong relationship between the tone, level of uncertainty and/or risk conveyed in US macro reports, and changes in the value of the VIX, on a quarterly basis. This suggests that analyst reports influence trading decisions and market behavior, and, therefore, can lead to self-fulfilling prophecies. Since those reports affect investor sentiment, they allow us to estimate future values of the VIX Index.

1. Introduction

Although widely used by professionals in the financial industry, the VIX Index is not as well-known by the public. Though historically it had been overlooked by retail investors because deemed too difficult to trade, this index became a trending topic at the beginning of the COVID-19 pandemic. Indeed, on March 20th, 2020, the VIX Index surpassed 66 points, reaching its highest value since the 2008 financial crisis. Suddenly, the VIX, which had been trading much lower (at around 10-15 points), and following a relatively stable trend, was being covered by all major financial news outlets, and discussed in trading chatrooms and forums across social media. Often misunderstood and even scrutinized [Griffin and Shams, 2017] the VIX gained a brand-new wave of popularity as markets underwent extreme volatility during the pandemic. Introduced by the CBOE in 1993, the VIX Index aims to measure market's expectation of 30-day volatility implied by S&P 500 options. Computed daily, is calculated by aggregating the weighted prices of S&P 500 puts and calls over the real-time mid-prices of their bid/ask quotes.

The VIX Index is calculated as:

$$VIX = 100\sqrt{\sigma^2}$$

where:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2$$

with:

T = time to expiration of the underlying index

ΔK_i = spacing between strike prices K_i

K_i = strike price of the i -th option

$Q(K_i)$ = mid price between Call Price and Put Price for each option with strike price K_i

F = forward index level

K_0 = first strike below the forward index level

R = risk-free interest rate

Intuitively, the implied variance consists of the expected variance of returns based on the options prices ($\frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i)$) minus a correction term to produce a risk-neutral variance estimate ($\frac{1}{T} [\frac{F}{K_0} - 1]^2$). Then, the VIX index assume that the Option prices contain market expectation of future volatility.

Commonly used as a proxy for investor sentiment by institutions across the globe, in less than twenty years, the VIX Index has become one of the most significant benchmarks for implied volatility on the US stock market. In 2006, CBOE also released the first VIX options contracts, allowing traders and investors to “hold” volatility. This became the exchange’s most popular product, as it provided pure volatility exposure, and therefore, made it possible to achieve greater diversification. It also gave rise to more research being conducted on how to predict changes in the VIX, given it could, from that point forward, be leveraged to create profit-maximizing strategies. Up to now, majority of the research has focused on predicting the VIX using quantitative metrics. [Baba and Sakurai, 2011] for instance relied on US macroeconomic data to forecast the index. More recently, [Osterrieder et al., 2020] used neural networks to predict the VIX based on a smaller set of options prices, that they identified as being the main driver of intraday changes in VIX values.

However, here, we wished to take a different approach, and predict the VIX Index based on textual information. Specifically, we will be looking at institutional research analysts’ US macro reports. Those reports have the advantage of incorporating macroeconomic data, while also conveying analyst sentiment. Research analysts play a key role in the financial ecosystem. Indeed, we often think of capital markets as being divided between the buy-side and the sell-side. On the buy-side are institutions representing individuals and/or companies with excess capital (ex: a pension fund). On the sell-side, on the other hand, are institutions representing individuals and/or companies seeking capital (ex: a bank). The role of buy-side market participants is thus

to find investment opportunities, while the role of sell-side market participants is to sell them. The most efficient way for sell-side market participants to sell these opportunities is through their reports. Read by buy-side investors every day, these reports not only provide ratings of major companies (for the case of company-specific reports), they also give market color (for the case of macro reports). We are interested in the latter, as these types of reports tend to be less biased. Because buy-side investors rely on them quite heavily, and actually use them to make trading decisions, we believe they influence market behavior. Then a pessimistic macro report may lead an investor to believe that markets are risky. If that is the case, equity options become more desirable: investors would rather have more room for flexibility and be shielded from downside risk. Thus, more investors would buy options, pushing their prices higher, and increasing the VIX Index. This can even happen if technically stocks are not getting any riskier, it suffices that enough analysts think that they are, for aggregated risk to increase. It is in that regard that we evoke the idea of **self-fulfilling prophecies**. Hence, we will assume that analyst sentiment influences the VIX index, and that this relationship is linear.

To achieve this analysis, we defined the following:

H_0 : The VIX Index moves independently

H_1 : Analyst sentiment influences the VIX Index

H_2 : There is a linear relationship between analyst sentiment and VIX Index

where H_0 represents the null hypothesis, H_1 and H_2 represents the alternative hypothesis.

2. Data

To conduct this analysis we will be using a database of over 35,000 online financial and economic news sources from 2017 to 2024. Articles were collected from multiple publicly available websites that regularly publish financial reports, market commentaries and macro-economic analyses. The scraping procedure yielded over 150,000 JSON files, each corresponding to a distinct article containing structured metadata fields such as title, full text, publication timestamp, sentiment, language, and topical categories.

To ensure the quality and relevance of the corpus, a Python-based extraction and filtering pipeline was developed. The program recursively traverses directories containing raw scraped data and consolidates all relevant articles into a single dataset for analysis. From each JSON file, only the essential fields — title, text, and publication date are retained, while other metadata are discarded to optimize processing speed and reduce noise in the subsequent analysis.

Given the objective of analyzing macroeconomic sentiment rather than firm-specific dynamics, we applied strict exclusion criteria to eliminate company-level reports. Articles containing explicit investment recommendations (e.g., “BUY”, “SELL”, “HOLD”) or references to corporate identifiers (such as Inc., Corp., Ltd., LLC) were removed. We also filtered out any reports mentioning common U.S. stock tickers (e.g., AAPL, MSFT, AMZN, NVDA, TSLA, etc.) to further isolate macroeconomic content from equity-specific discussions.

To ensure thematic consistency, we retained only articles classified under broad macroeconomic categories, including “Economy, Business and Finance”, “Macroeconomics”, “Fiscal Policy”, “Monetary Policy”, “Inflation”, “Markets”, and “Global Trade”. The resulting dataset thus focuses on articles that reflect aggregate economic developments, policy discussions, and market-wide sentiment rather than individual firm news.

After the data cleaning, we began performing topic modeling analysis on the data, to get a better sense of what the main trends were during this period. To do so, we used Latent Dirichlet Allocation (LDA) [project, 2025]. This technique consists in sorting a set of reports into different topics. A topic is determined by deriving what the top ten words are across the reports. For this analysis we limited ourselves to five topics. Then, for each word of each report, we compute the probability that this word belongs to topic 1, topic 2 etc. . . . Our analysis revealed that the dominant topics during our period (2017-2024) were: Growth Stock and Crypto, Fiscal Policy, Consumer Trends, Economic Policy, Inflation

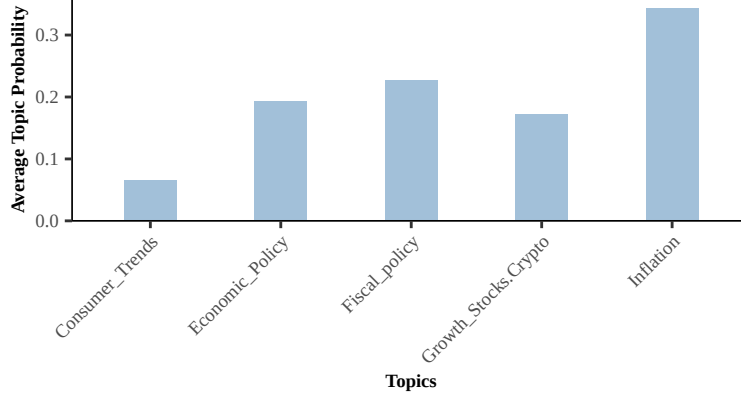


Figure 1: Distribution of Dominant Topics from LDA Analysis

We note that inflation emerges as the dominant topic, significantly higher than other macroeconomic themes. This prominence is particularly relevant given inflation’s direct impact on market volatility through monetary policy response.

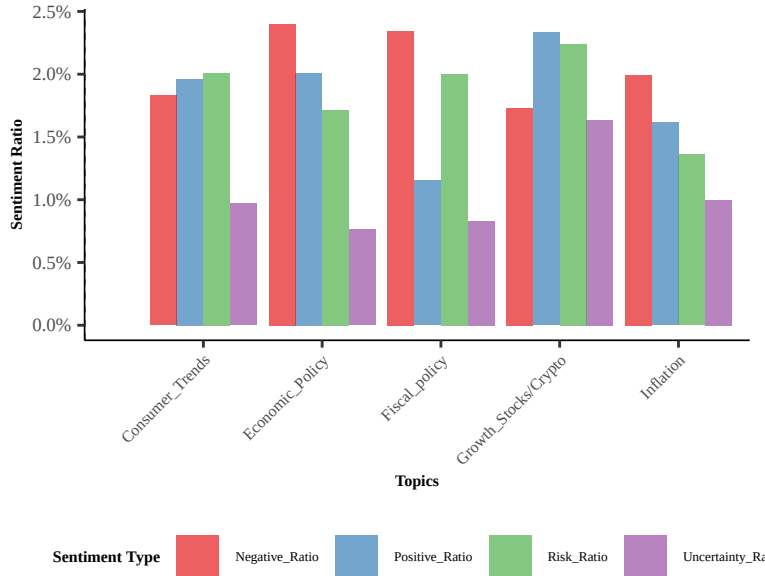


Figure 2: Sentiment Ratios by Topic Using Loughran-McDonald Dictionary

Figure 2 reveals several key sentiment trends across macroeconomic topics. Overall, the discourse exhibits a cautious tone, with most topics showing higher negative than positive sentiment. Growth Stocks/Crypto stands out as the notable exception, displaying the most optimistic coverage with the highest positive sentiment ratio. Fiscal Policy emerges as the most negatively framed topic, likely reflecting contentious debates around government spending. The consistently elevated risk and uncertainty ratios across all topics suggest a pervasive awareness of economic vulnerabilities, with Growth Stocks/Crypto showing particularly high uncertainty, consistent with the volatile nature of these assets.

The last piece of data that we need is the VIX Index data. This data was taken from Yahoo Finance, averaged into quarters and limited to adjusted close prices only, to account for any reporting discrepancies. Having gathered, cleaned, and analyzed our data, we are now ready to model it.

3. Model

In this section of the reports, we will be testing our hypotheses by building a model to explain changes in the VIX Index.

Our model includes the following explanatory variables:

- x_1 : Ratio of words that express risk
- x_2 : Ratio of words that express uncertainty
- x_3 : Ratio of negative words
- x_4 : Ratio of positive words

As explained prior, we believed these variables are representative of overall analyst sentiment, which we assume impact the beliefs held by the market participants regarding aggregated risk. These ratios were computed using the Loughran-McDonald Master Dictionary w/ Sentiment Word Lists (2024) [University of Notre Dame, 2024], which contains lists of words that express the dimensions of sentiment we have identified, mainly tone, risk and uncertainty.

We created a program that counted such words, and then divided that count by the total number of words present in all the reports for that quarter. Ratio allowed us to account for differences in the length of reports. The output of this program was a consolidated csv file that included sentiment ratios for all quarters in our set, as well as the corresponding VIX index values. We then loaded this data onto Rstudio and proceeded to split it into a training and testing set. We will be testing out 2 alternative hypotheses (versus the null):

H_1 : Analyst sentiment influences the VIX Index

H_2 : There is a linear relationship between analyst sentiment and the VIX Index

We began the modelling process by running a stepwise selection algorithm on the training data to identify the best linear model as efficient as possible. Indeed, in stepwise selection, variables are added to the model sequentially, and removed whenever they no longer provide improvement in the model fit. In that sense, stepwise selection is a hybrid between forward and backward selection, with major computational advantages as compared to best-subset selection, which performs an exhaustive search.

This model, that we will be referring to as Model0, will also serve as a benchmark throughout our analysis. This is, in a sense, the “naïve” model, that assumes linearity and gaussian noise. It is the simplest and most accurate model we could have come up with given our four explanatory variables. The output of the stepwise selection algorithm reveals the best fit given our variables is:

$$VIX = \beta_0 + \beta_1 X_1 + \beta_2 X_4 + \epsilon$$

This model has a p-value < 0.05 , then we can reject the null hypothesis, stating that the VIX moves independently from analyst sentiment, for this level of significance, X_4 , ratio of positive words, is also significant at the 5% level and although X_1 , ratio of words that express risk is not, it appears to be adding predictive power to the model.

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Call:
lm(formula = Y ~ X1 + X4)

Residuals:
    Min       1Q   Median       3Q      Max
-8.3712 -3.2859 -0.8539  1.9868 13.7518

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    32.59      13.85   2.353  0.02757 *
X1             3249.01    1191.13   2.728  0.01200 *
X4            -2629.91     757.79  -3.470  0.00207 **
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.152 on 23 degrees of freedom
Multiple R-squared:  0.5637,    Adjusted R-squared:  0.5258
F-statistic: 14.86 on 2 and 23 DF,  p-value: 7.205e-05

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Figure 3: Model0 Regression Output

We note the models key statistics:

Table 1: Model Performance Metrics

Model	R ²	Adjusted R ²	OOS R ²	CP	AIC	BIC
Model 0	0.5637	0.5258	0.3064	48.7049	163.8446	168.877

This allows us to validate H_1 : there is indeed a relationship between the VIX and analyst sentiment. However, although Model0 performs somewhat well in sample, out-of-sample, its predictive power is considerably lower. Its CP, AIC and BIC are also quite high indicating this might not be the best fit for the data. This prompts us to the question of the validity of the linearity hypothesis. We can see the generated fitted vs residual graph (see figure 4) and QQ-plot (see figure 5) exhibit non-linear trends. Moreover, the pairwise scatterplots reveal linear correlations between some of the explanatory variables, indicating that including interaction terms might improve the fit. For instance, we observe a somewhat strong linear correlation between X_1 and X_3 , respectively ratio of words that express risk and ratio of negative words. This make sense, as we can imagine that reports that have a higher count of words that express risk also have a more pessimistic tone, since risk is negatively connotated, especially in a context where market participants are risk averse.

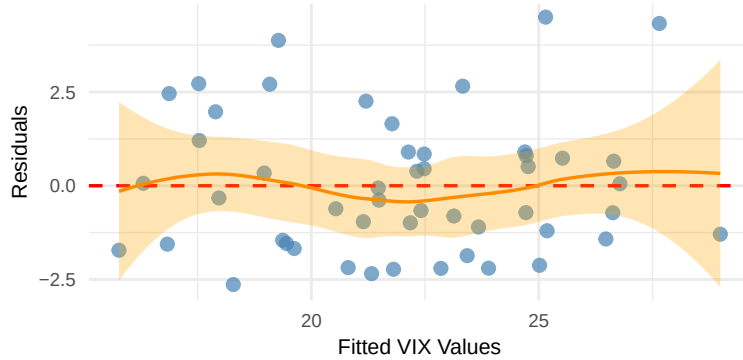


Figure 4: Residuals vs Fitted Values - Model 0

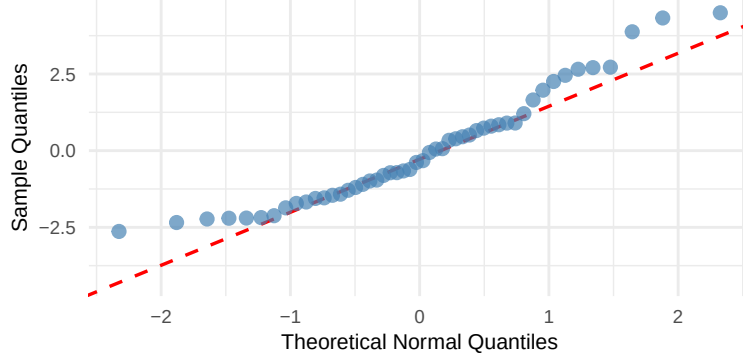


Figure 5: Model0 QQ-plot

As we can see, a large majority of these non-linear models have a better adjusted R^2 than model0, the best linear model for the data. Comparing key performance indicators between Model0 and the top five non-linear models yields:

Table 2: Comparison of Linear vs Non-Linear Models Performance

	Model	R^2	Adjusted R^2	OOS R^2	CP	AIC	BIC
2	Model1	0.8919	0.8577	0.7827	5.0250	135.5711	145.6358
4	Model3	0.9027	0.8569	0.7574	7.3191	136.8259	149.4069
3	Model2	0.8963	0.8560	0.7716	6.3308	136.4888	147.8116
5	Model4	0.9047	0.8510	0.7350	9.0135	138.3021	152.1411
6	Model5	0.9047	0.8412	0.6680	11.0000	140.2786	155.3758
1	Model0	0.5637	0.5258	0.3064	48.7049	163.8446	168.8770

We can see that Model1 is the most relevant, with the highest adjusted R^2 , out-of-sample R^2 , Mallows CP, AIC and BIC. Its R^2 is slightly lower than the R^2 of other non-linear models, but that is to be expected, since it is the model with the least number of predictors among the non-linear ones. Overall, non-linear models seem to be more accurate, with a significant increase in adjusted R^2 , out-of-sample R^2 , and much lower CP and AIC values. We especially note that the out-of-sample R^2 for Model1 is almost double that of Model0. We conclude that Model1 is the best model. According to model1:

$$VIX = \beta_0 + \beta_1 X_2 + \beta_2 X_1 X_2 + \beta_3 X_1 X_3 + \beta_4 X_2 X_3 + \beta_5 X_2 X_3 + \epsilon$$

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Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.921e+02  5.349e+01  -5.461 2.87e-05 ***
X2           4.486e+04  7.457e+03   6.015 8.70e-06 ***
X2:X1       -2.171e+06  9.039e+05  -2.401 0.02673 *
X1:X3        2.787e+06  4.191e+05   6.649 2.33e-06 ***
X1:X4       -2.256e+06  6.277e+05  -3.595 0.00193 **
X2:X3       -1.487e+06  2.175e+05  -6.840 1.58e-06 ***
X3:X4        4.541e+05  1.737e+05   2.615 0.01705 *
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.822 on 19 degrees of freedom
Multiple R-squared:  0.8919,    Adjusted R-squared:  0.8577
F-statistic: 26.12 on 6 and 19 DF,  p-value: 3.262e-08

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Figure 6: Model1 Regression Output

From figure 6, we can see that all predictors except the X_3X_4 and X_2X_1 interaction term are significant at the 0.05 level. The p-value is also < 0.05 which means that, once again, we can reject the null hypothesis.

Testing model1 on the remaining 40% of the data (testing set), we have the following results:

Table 3: Model1 Output on testing Data

Model	R^2	Adjusted R^2	RMSE
Model 1	0.8919	0.8577	2.4862

Even on new data, we still gave both a high R^2 and adjusted R^2 . Model1 is able to explain more than 85% of the variation in the VIX Index. The low root-mean square error value also suggests strong predictive power. Figure 7 illustrates the accuracy of Model1.

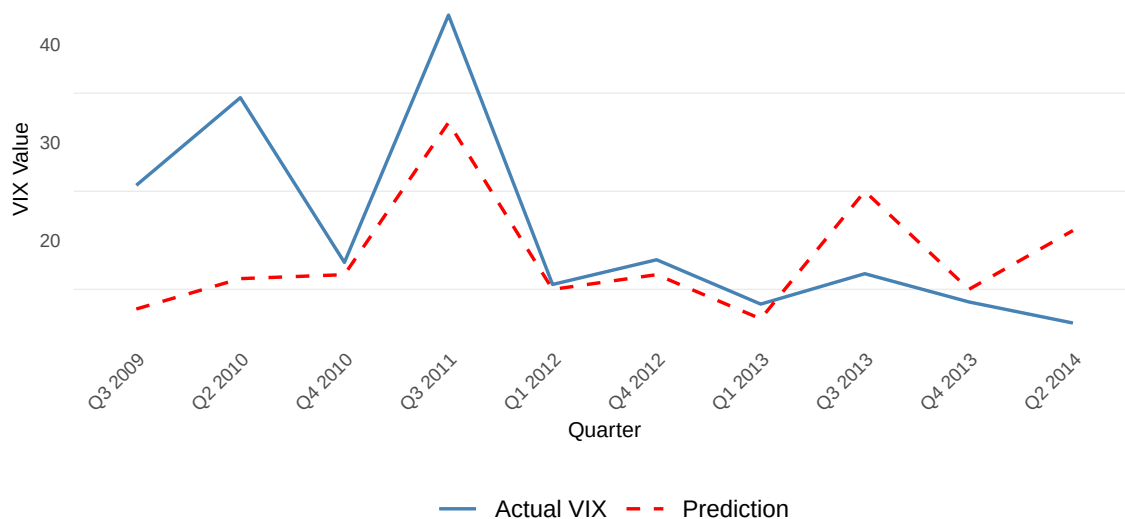


Figure 7: VIX Actual vs Predicted Values

4. Conclusion

This analysis revealed the effect of institutional analysts' macro reports on the VIX Index. Although often modeled by relying solely on quantitative metrics, our work focused on predicting the VIX using textual information. We found that the bulk of macro reports are heavily skewed towards themes and topics that directly relate to market volatility (inflation, growth stocks, the volatility curve), suggesting analysts will often give an opinion on overall market risk. Because of their importance within the financial ecosystem, we conjectured that their views would directly impact trading decisions, and thus, indirectly impact the VIX Index. Investors do pay close attention to the messages conveyed in research reports, especially those conveyed in macro reports, as macro reports tend to be less biased. Hence, we wanted to test the hypothesis that analyst sentiment, because it influences investor sentiment, can help us gauge expected market volatility. After modelling the data using a multiple linear regression model, we indeed found a relationship between analyst sentiment, as captured by tone (ratio of positive words and ratio of negative words), level of risk conveyed (ratio of words that express risk), level of uncertainty conveyed (ratio of words that express uncertainty) and the VIX Index. However, a linear model did not appear to be the best fit for the data, as it exhibited non-linear trends. We also found that certain explanatory variables (like ratio of words that express risk and ratio of negative words) were linearly correlated, indicating interactions between terms. Thus, we ran a non-linear model and found that this model, which incorporated interaction terms, could explain around 85%

of the variation in the VIX Index. Allowing for the joint effect of predictors on the VIX Index significantly improved the model’s predictive power, both in-sample and out-of-sample. According to this model, the VIX is especially sensitive to reports that express uncertainty (X_2 in the regression output, see figure 6), which makes sense as options are priced at their highest during uncertain times. This is what we have seen throughout the COVID-19 pandemic. Most interaction terms in the model are also significant at the 5% level, suggesting they are necessary in the regression. We conclude that aggregate analyst sentiment is a good predictor of the VIX Index. Although the model is not always accurate, it performs well in terms of predicting the direction in which the VIX will move, which is crucial both for creating profit-maximizing strategies and for gauging investor fears. Being able to predict the VIX is also helpful in predicting returns of the S&P 500 [Fernandes et al., 2014], as there exists a negative correlation between the VIX Index and S&P 500 returns. Overall, this index not only reveals a lot of information on market participants’ views on future volatility, it also helps explain stock market returns. Although sometimes overlooked, there is plenty of value in being able to predict the VIX Index.

Despite being confident in our results, we note a few limitations of our analysis: adding complexity to the model proved successful and efficient in our case, since we were dealing with a relatively small dataset, however, had we transformed the data beforehand (by taking its log for instance), we would have avoided the need to complexify the regression. This would have been particularly useful if we had been working with a much larger dataset, and with more predictors, as adding more interaction terms would have made the model more difficult to interpret. Capturing sentiment is also a challenging component of this analysis: we only relied on three main dimensions (tone, risk and uncertainty) but we could have included more (extent to which the report is forward-looking, extent to which hyperbolic language is used etc. . .). We also could have included a transparency component to the analysis, and evaluated the quality of the information itself. One could conjecture that longer, more detailed reports are likely to be about risk increasing rather than decreasing. Lastly, it is important to recognize the limitations of the natural language processing techniques used, as they are not yet completely accurate, even on large data samples.

References

- N. Baba and Y. Sakurai. Predicting regime switches in the VIX index with macroeconomic variables. *Applied Economics Letters*, 18(15):1415–1419, 2011.
- Marcelo Fernandes, Marcelo C. Medeiros, and Marcel Scharth. Modeling and predicting the CBOE market volatility index. *Journal of Banking & Finance*, 40:1–10, 2014.
- John M. Griffin and Amin Shams. Is the VIX a good predictor of market volatility? *Journal of Finance*, 72(3):1235–1276, 2017.
- Joerg Osterrieder, Daniel Kucharczyk, Silas Rudolf, and Daniel Wittwer. Neural networks and arbitrage in the VIX: A deep learning approach for the VIX. *Digital Finance*, 2(1):97–115, 2020.
- Contributing project. Topic modeling with latent dirichlet allocation, 2025. URL <https://github.com/lda-project/lda?tab=contributing-ov-file>. Example of LDA application on Reuters dataset.
- University of Notre Dame. Loughran-McDonald master dictionary with sentiment word lists, 2024. URL <https://sraf.nd.edu/loughranmcdonald-master-dictionary/>. Software repository for accounting and finance.